Consumer welfare beyond GDP

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Abstract

I construct welfare when consumers have heterogeneous preferences, unequal incomes, and face different prices. The theory builds on three axioms: collective welfare (i) is maximized at *Laissez-faire* when consumers have identical budgets, (ii) treats same-preference consumers anonymously, (iii) is separable across economies. I prove a new incompatibility between them whenever there is price dispersion. Then, possibilities are obtained when separability only applies across economies with identical aggregate resources. As standard, these possibilities include a sum of expenditure functions evaluated at a common price. The main novelty is that they also characterize the common price as an income-weighted harmonic mean of individual prices. I apply these results to the measurement of costs and standards of living. For international comparisons of real GDP, one must adopt modified PPP indices that can be made robust to preference heterogeneity without additional data. For growth and inflation measurement, one must adopt constant base prices reflecting all available years.

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1 Introduction

When consumers have heterogeneous preferences and budgets, measuring individual and collective consumer welfare requires answers to three distinct questions: (i) how to compare consumers interpersonally, (ii) how to deflate their expenditures from different prices, and (iii) how to aggregate individual welfare into collective welfare. For example, the Penn World Table[1] deflates each country's consumption by common "world" prices to compare real income across countries. Deaton and Heston (2010) stressed that this practice (i) neglects preference heterogeneity, (ii) uses plutocratic world prices which are influenced more by richer countries, and (iii) ignores inequalities.

The present paper proposes an integrated theory that compares the welfare of consumers both collectively and interpersonally while informing on deflation practices. I build axiomatically a measure of consumer welfare in a general consumption space where agents may have heterogeneous preferences, endowments, and personalized prices.

The theory uses three kinds of axioms. The first axiom imposes that, if all agents had the same budgets, letting them choose their preferred bundle in that budget, i.e. *Laissez-faire*, yields an allocation better than any other feasible allocations. The second axiom requires that if two agents have identical preferences but unequal budgets, swapping their consumption bundles does not affect social welfare. The third axiom is the standard Fleming (1952) Separability[2] whereby indifferent agents do not matter, changing their budgets, preferences, or consumption does not affect social welfare as long as they remain indifferent.

These axioms are compatible with any degree of aversion to inequality. They nest the case of equality of opportunity, where the observer is averse to consumption inequalities due to unequal budget but not averse to consumption inequalities due to heterogeneous preferences. As such, the approach resonates with the debate on building economic indicators *beyond GDP* (Fleurbaey, 2009; Fleurbaey & Blanchet, 2013).

The first main result shows that these axioms are incompatible when there is price dispersion. To see why, consider two different economies where all agents have identical budgets. Because both prices and income in the first economy may be different from the ones in the second economy, it may be that some allocations are feasible in both of them. Then, we can construct cases where the Laissez-faire allocation of the first economy is feasible in the second, and *vice versa*. This is enough to engender impossibilities: the first axiom ranks the Laissez-faire allocation as optimal in the first economy and the other axioms imply a stability of that ranking between the two economies, but the first axiom ranks another allocation as optimal in the second economy.

The paper argues that one should escape the impossibility by weakening the third axiom.

[1]See Feenstra, Inklaar, and Timmer (2015) and Summers and Heston (1991) for the construction of the Penn World Table.

[2]Fleming (1952) proved that this Separability axiom, combined with Strong Pareto, leads to a utilitarian social welfare function

Intuitively, modifying agents' income and prices is not innocuous because it directly affects what is feasible in the economy, which matters through the first axiom. I propose to apply Separability between two economies only if they share the same quantity of resources in each good. This is enough to escape the impossibility. I prove several representation theorems of social welfare functions that can be used to compare the welfare of consumers interpersonally and collectively.

The second main result of this paper is that the individual consumer welfare measure should be an expenditure function that uses an identical price vector for all consumers. This common price vector reflects the purchasing power of the economy: the price of good l is the ratio between the economy's aggregate income and the total quantity of good l in resources. Equivalently, the common price for a good is the income-weighted harmonic mean of individual prices.

The third set of results aggregates this individual consumer welfare into collective consumer welfare. I use known axiomatic results and show that the aggregator can be a sum or a maximin operator. Interestingly, the degree of inequality aversion in the social welfare function follows a one-to-one relationship with the observer's inequality aversion between consumers with identical prices. This suggests that welfare must be constructed in a two-step procedure: first, compare interpersonally heterogeneous consumers using their expenditure function with the common price; second, apply the same inequality aversion between them as you would if there was no price dispersion.

These results provide welfare foundations for a large set of cost and standards of living measurement problems. For international comparisons of real GDP, the paper recommends adopting a different Purchasing Power Parity (PPP) index than what is typically done. While the standard Geary-Khamis international dollar is based on a consumption-weighted harmonic mean of countries' prices, this paper argues for substituting consumption weights with income weights. Indeed, this new PPP index is robust to preference heterogeneity because it is invariant to preference shocks conditional on income and prices, while consumption weights are not. Most importantly, it does not require more data than what is collected by statistical agencies.[3] This also answers Deaton and Heston (2010) concerns: PPP indices (i) can be robust to preference heterogeneity at little cost, (ii) can be plutocratic without conflicting with fairness, and (iii) do not depend on the observer's inequality aversion. Finally, I also show the implications of the paper for the problem of measuring inflation and growth in one country over time. In that context, the paper provides a foundation for constructing a base year that is consistent with welfare.

Literature

Comparing agents with heterogeneous preferences is notoriously difficult since Arrow (1950)'s impossibility theorem. Indeed, there is no obvious way of summing up utilities when utility

[3] This has consequences not only for real GDP but also for its derived statistics to measure labor productivity, poverty, or inequality as in the Distributional National Accounts as in Piketty, Saez, and Zucman (2018).

functions differ across individuals. The present paper escapes Arrow's impossibility by using the social choice theory of fairness initiated by Fleurbaey and Maniquet (2006, 2011, 2018). It uses information on indifference curves, while Arrow's theorem only used utility levels. This allows us to characterize welfare measures in the realm of *equivalent income*, such as willingness to pay, expenditure functions, or money-metric utility functions. For example, the Hicksian compensating and equivalent variations belong to this class.

A lasting[4] critique of these expenditure functions is that they depend on a reference price that is left at the researcher's discretion (see discussions in e.g. Bosmans, Decancq, and Ooghe (2018), Capéau, Decoster, and De Sadeleer (2023), and Fleurbaey and Blanchet (2013)). This limitation motivated the study of reference-independent welfare assessment (Blackorby, Laisney, & Schmachtenberg, 1993; Eden & Freitas, 2024; Roberts, 1980). The main contribution of this paper with respect to the social choice literature is to axiomatize welfare measures based on expenditure functions whose reference price is endogenized by the axioms. This comes from the fact that I consider price dispersion across individuals, while previous papers imposed homogeneous prices for all agents, building on the Walrasian tradition in general equilibrium theory. Yet, the law of one price does not hold in many datasets.[5] This papers show that this extra dimension of heterogeneity engenders an impossibility, which in turn leads to new characterization results.

This paper also contributes to the index number theory as initiated by Fisher (1922). The index number problem consists in building price and quantity indices based on dataset of observed prices and expenditures for each consumer. van Veelen and van der Weide (2008) divide the proposed solutions in two approaches. The axiomatic approach imposes series of desirable mathematical properties on the indices but is typically not linked with classical demand theory (Samuelson & Swamy, 1974; Van Veelen, 2002). The economic approach rationalizes observed differences from a representative consumer demand system (Diewert, 1976; Neary, 2004). The present paper rejects a representative consumer unlike the latter but has links with classical demand theory unlike the former. Moreover, the paper does not impose any domain restrictions on individual preferences, such that its results are valid whether preferences are homothetic (Diewert & Nakamura, 1993) or not (Jaravel & Lashkari, 2024).

The paper is structured as follows. In section 2, I present the standard consumption space and define the objects of interest. In section 3, I present the key axioms and prove their incompatibilities. In section 4, I weaken separability and derive several representation theorems. In section 5, I derive implications for various applications. In section 6, I conclude.

[4]A classical critique by Blackorby and Donaldson (1988) was that money-metric utility function may lead to inegalitarian assessment. However, Fleurbaey and Maniquet (2011) and Schlee and Khan (2022, 2023) showed multiple ways to have inequality-averse social welfare functions based on money-metric utility.

[5]For example, the Penn effect describes the fact that non-tradable goods are more expensive in richer countries, as documented by Kravis, Heston, and Summers (1978) and explained in trade theory by Balassa (1964) and Samuelson (1964) (see Samuelson (1994) for a review).

2 Model

The environment is a standard consumption space. A society $\mathcal{N} = \{1, ..., N\}$ is composed of a finite number N of consumers. There are $L \geq 2$ private goods and a consumer i's consumption bundle is denoted by $z_i \in \mathbb{R}^L_+$.

Agents are heterogeneous in their preference ordering $\succsim_i$ which is assumed to be complete, transitive, locally non-satiated, and convex, as well as in their endowments in goods[6] $\omega_i \in \mathbb{R}^L_+$ and personalized prices $p_i \in \mathbb{R}^L_+$. Personalized prices may include the standard case of no price dispersion in a Walrasian equilibrium, as well as price dispersion due to e.g. search frictions, transport costs, price discrimination, regulations, heterogeneous needs, or comparative advantages in production. For brevity, I will refer to the scalar product $p_i \cdot \omega_i$ as income and denote it by $y_i = p_i \cdot \omega_i \geq 0$.

An allocation z is a list of consumption bundles for each agent $z = (z_1, z_2, ..., z_N) \in \mathbb{R}^{LN}_+$ and it may or may not be rationalizable by agents' utility maximization constrained by a budget set. An economy e is composed of N agents which are heterogeneous in preferences, endowments, and prices $e = \{(\succsim_i, \omega_i, p_i)_{i \in \mathcal{N}}\}$. The set of all economies is denoted by $\mathcal{E}$.

For each economy e, the observer has a complete and transitive ordering **R(e)** over allocations $z \in \mathbb{R}^{LN}_+$. This will be the main object of interest of the paper. I use **R(e)**, **P(e)**, **I(e)** for weak preference, strict preference, and indifference, respectively.

Because social welfare is constructed in the first-best, the observer can use lump-sum transfers. It will be useful to define the following two concepts. An allocation $\hat{z}$ is feasible by lump-sum transfers from the economy e if there exists a vector of individual lump-sum transfers $(t_1, t_2, ..., t_N) \in \mathbb{R}^N$ with $\sum_{i \in \mathcal{N}} t_i \leq 0$ such that bundles $\hat{z}_i$ are part of individuals' budgets $B(t_i, \omega_i, p_i)$:

$$\hat{z}_i \in B(t_i, \omega_i, p_i) = \{z_i \in \mathbb{R}^L_+ : p_i \cdot z_i \leq p_i \cdot \omega_i + t_i\}$$

These allocations $\hat{z}$ are objects in $\mathbb{R}^{LN}_+$. For geometric intuitions below, it will be useful to define a similar concept in $\mathbb{R}^L_+$. The aggregate budget of an economy, denoted by $\mathcal{B}(e)$ is the set that collects all the consumption vectors that can be consumed given the economy's resources. Formally, it is obtained as the (Minkowski) sum of individual budgets[7], i.e.

$$\mathcal{B}(e) = \{\sum_{i \in \mathcal{N}} z_i \in \mathbb{R}^L_+ : z_i \in B(0, \omega_i, p_i) \quad \forall i \in \mathcal{N}\}$$

Thus, an allocation $\hat{z} \in \mathbb{R}^{LN}_+$ is feasible by lump-sum transfers from economy e if and only if the sum of bundles across individuals belongs to the aggregate budget of that economy $\sum_{i \in \mathcal{N}} \hat{z}_i \in \mathcal{B}(e)$.

[6]Observe that this model nests a model where one good is set as the numéraire and endowments are only in that good, that is a monetary endowment. All our results are independent of the choice of the numéraire.

[7]While Minkowski summation preserves convexity, it does not necessarily preserve linearity. In general, $\mathcal{B}(e)$ will be a piecewise-linear convex set and it coincides with the simplex if and only if $p_i = p_j$ for all $i, j \in \mathcal{N}$.

3 Impossibilities

The axiomatic construction of the social ordering **R(e)** consists of imposing desirable properties, i.e., axioms, which restrict the scope of possibilities among the universe of transitive and complete **R(e)**. This section introduces the key axioms of the paper and shows that they lead to an impossibility.

The first axiom, **Laissez-faire**, embodies the view that agents deserve the consequences of their tastes.[8] Formally, it requires that when there is no cross-sectional heterogeneity in prices and endowments, the Laissez-faire allocation, where each agent freely chooses what they prefer out of their budget set $B(0, \omega, p)$, is strictly preferred to any other allocations feasible by lump-sum transfers from the economy.

Laissez-faire: $\forall e \in \mathcal{E}$ such that $(\omega_i, p_i) = (\omega, p)$ for all $i \in \mathcal{N}$, we have

$$z^* \ \ \mathbf{P(e)} \ \ z$$

for z^* such that $z_i^* \in \max_{\succsim_i} B(0, \omega, p)$ for all $i \in \mathcal{N}$ and any $z \neq z^*$ feasible by lump-sum transfer for e.

In the context of consumer welfare measurement, this axiom implies that, when everyone has the same budget, welfare should be constructed based on an agent's own favorite bundle rather than any other bundle. For example, this axiom precludes the economist from choosing an arbitrary bundle and evaluating distances to it as welfare, such as a poverty bundle or a given CPI consumer basket.

In and of itself, this axiom does not force the planner to be inequality averse. However, for inequality-averse planners, it imposes that the part of consumption inequalities that are due to heterogeneous tastes are unproblematic. This means that interpersonal inequalities in welfare may not be driven by heterogeneous preferences. Relatedly, a resource allocation which fully equalizes consumption is not optimal if this axiom is imposed because part of consumption inequalities should not be reduced as they reflect unequal tastes.

The second axiom, **Same-Preference Anonymity**, reflects the view that agents do not deserve the consequences of their budgets. In particular, the axiom imposes that when a pair of agents have identical preferences, a permutation of their bundles should be socially indifferent.[9] In other words, the social ranking is anonymous among agents with identical preferences.

[8] Political philosophers such as Arneson (1990), Cohen (1989), Dworkin (1981), and Rawls (1971) all defended variants of this principle, as discussed in Fleurbaey (2008) and Fleurbaey and Maniquet (2006, 2011).

[9] This type of axioms also received support in political philosophy among liberal egalitarians (Dworkin, 1981). It is reminiscent to Suppes (1966)'s grading principle. See Hammond (1976) for an early treatment and Maniquet (2004) for its relationship to equality of opportunity.

Same-Preference Anonymity: $\forall e \in \mathcal{E}$ where agents i, j are such that $\succsim_i = \succsim_j$ then

$$(z_i, z_j, z_{\mathcal{N}\setminus\{i,j\}})\mathbf{I(e)}(z_j, z_i, z_{\mathcal{N}\setminus\{i,j\}}).$$

This axiom introduces a mild degree of compensation in the social welfare function. Indeed, **Same-Preference Anonymity** says that swapping the bundles from agents with large budgets to agents with small budgets does not change social welfare, but neither does the opposite. As a consequence, agents do not deserve the consequences of their budgets. Equivalently, welfare should be constructed in the same way for all agents with the same preferences. In the context of measuring consumer welfare across countries, this axiom forbids comparing equal-preference countries with country-specific measuring rods.

Taken together, **Laissez-faire** and **Same-Preference Anonymity** do not determine how much inequality should society dislike, but they rather restrict the information that can be used when constructing welfare. If these two axioms were combined with an inequality-aversion requirement, they would lead to a social welfare function that reflects the ethics of equality of opportunity: consumption inequalities arising from unequal budgets should be reduced, but consumption inequalities arising from unequal preferences should not.

Observe that both axioms only apply under some specific premises, i.e. for some but not all e in $\mathcal{E}$. In order to obtain a complete ordering **R(e)** for all economies, one must use a cross-economy robustness condition. Since the characterization of utilitarianism by Fleming (1952), the standard cross-economy condition requires that indifferent agents should not have a say in the social evaluation between two alternatives. This is sometimes defended on democratic grounds: only those effectively affected by the alternatives should be heard. More formally, **Fleming (1952) Separability** requires that if some agents $\mathcal{K}$ are receiving the same bundle in two allocations, then the social ordering between these two allocations is unaffected if these agents change their preferences, endowments, prices, or receive a different bundle.

Fleming (1952) Separability: $\forall e, e' \in \mathcal{E}$ and the partitions $\mathcal{K} + \mathcal{M} = \mathcal{N}$ with $e_{\mathcal{K}} = \{(\succsim_i, \omega_i, p_i)_{i\in\mathcal{K}}\}$, $e'_{\mathcal{K}} = \{(\succsim'_i, \omega'_i, p'_i)_{i\in\mathcal{K}}\}$, $e_{\mathcal{M}} = \{(\succsim_i, \omega_i, p_i)_{i\in\mathcal{M}}\} = e'_{\mathcal{M}}$, and z, z' then,

$$(z_{\mathcal{K}}, z_{\mathcal{M}})\mathbf{R}(\mathbf{e}_{\mathcal{K}},\mathbf{e}_{\mathcal{M}})(z_{\mathcal{K}}, z'_{\mathcal{M}}) \iff (z'_{\mathcal{K}}, z_{\mathcal{M}})\mathbf{R}(\mathbf{e}'_{\mathcal{K}},\mathbf{e}_{\mathcal{M}})(z'_{\mathcal{K}}, z'_{\mathcal{M}})$$

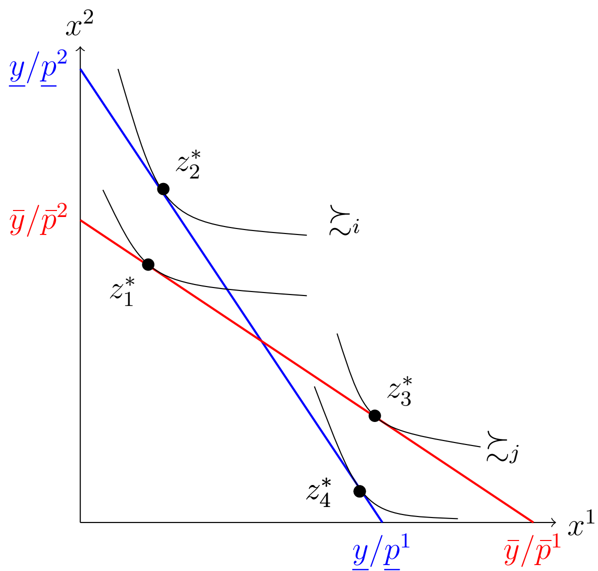
I note that the version presented here is logically weaker than the axioms of separability in subpopulations used in d'Aspremont and Gevers (1977), Fleurbaey (2003), and Maniquet (2004). The next theorem shows that the three axioms outlined above lead to an impossibility when combined together.

Theorem 1. *There does not exist any **R(e)** that satisfies **Laissez-faire, Same-Preference Anonymity** and **Fleming (1952) Separability***

Proof. It is enough to prove the statement when $N = 4$. By contradiction, assume that the statement does not hold. We will work with economies constructed from two preference profiles $\succsim_i$ and $\succsim_j$ and two Laissez-faire budgets $B(0, \bar{\omega}, \bar{p})$ and $B(0, \underline{\omega}, \underline{p})$. There are four bundles of interest that maximize either $\succsim_i$ or $\succsim_j$ over the budgets $B(0, \bar{\omega}, \bar{p})$ and $B(0, \underline{\omega}, \underline{p})$

and we denote them by $z_1^*, z_2^*, z_3^*, z_4^*$ respectively. Moreover, preferences and budgets are chosen such that $z_1^*, z_2^*, z_3^*, z_4^*$ are the vertices of a parallelogram. The situation is depicted in Figure 1.

Figure 1: Illustration of the proof of Theorem 1 where $\bar{y} = \bar{p}\bar{\omega}$ and $\underline{y} = \underline{p}\underline{\omega}$.



Notes: The quantity of good 1 is denoted by x_1 on the horizontal axis and the quantity of good 2 is denoted by x_2 on the vertical axis. The economies have been chosen such that the Laissez-faire allocation in the blue economy is feasible as lump sum transfer from the red economy, and *vice versa*, that is $\frac{z_1^*+z_3^*}{2} = \frac{z_2^*+z_4^*}{2}$.

Geometrically, the diagonals of a parallelogram bisect each other, such that $\frac{z_1^*+z_3^*}{2} = \frac{z_2^*+z_4^*}{2}$. This implies that both Laissez-faire allocations use an identical amount of aggregate resources

$$z_1^* + z_1^* + z_3^* + z_3^* = z_2^* + z_2^* + z_4^* + z_4^*$$

Hence, $(z_1^*, z_1^*, z_3^*, z_3^*)$ is feasible as lump-sum transfers from economy where all agents have the blue budget, and $(z_2^*, z_2^*, z_4^*, z_4^*)$ is feasible as lump-sum transfers from the economy where all agents have the red budget.

By **Laissez-faire**,

$$(z_1^*, z_1^*, z_3^*, z_3^*) \qquad \mathbf{P}\left(\left\{(\succsim_i, \bar{\omega}, \bar{p}), (\succsim_i, \bar{\omega}, \bar{p}), (\succsim_j, \bar{\omega}, \bar{p}), (\succsim_j, \bar{\omega}, \bar{p})\right\}\right) \qquad (z_2^*, z_1^*, z_3^*, z_4^*)$$

By **Fleming (1952) Separability**,

$$(z_1^*, z_2^*, z_4^*, z_3^*) \qquad \mathbf{P}\left(\left\{(\succsim_i, \bar{\omega}, \bar{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \bar{\omega}, \bar{p})\right\}\right) \qquad (z_2^*, z_2^*, z_4^*, z_4^*)$$

By **Same-Preference Anonymity**,

$$(z_2^*, z_1^*, z_3^*, z_4^*) \qquad \mathbf{I}\left(\left\{(\succsim_i, \bar{\omega}, \bar{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \bar{\omega}, \bar{p})\right\}\right) \qquad (z_1^*, z_2^*, z_4^*, z_3^*)$$

By Transitivity,

$$(z_2^*, z_1^*, z_3^*, z_4^*) \qquad \mathbf{P}\left(\left\{(\succsim_i, \bar{\omega}, \bar{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \bar{\omega}, \bar{p})\right\}\right) \qquad (z_2^*, z_2^*, z_4^*, z_4^*)$$

By **Fleming (1952) Separability**,

$$(z_2^*, z_1^*, z_3^*, z_4^*) \qquad \mathbf{P}\left(\left\{(\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p})\right\}\right) \qquad (z_2^*, z_2^*, z_4^*, z_4^*)$$

But this contradicts **Laissez-faire**, which completes the proof. ■

A natural attempt to escape the impossibility would consist in weakening **Laissez-faire** to a weak social preference for the Laissez-faire allocation rather than a strict preference. This is what the next axiom, **Laissez-faire**^{W}, achieves.

Laissez-faire^W: $\forall e \in \mathcal{E}$ such that $(\omega_i, p_i) = (\omega, p)$ for all $i \in \mathcal{N}$, we have z^***R(e)**z for z^* such that $z_i^* \in \max_{\succsim_i} B(0, \omega, p)$ for all $i \in \mathcal{N}$ and any $z \neq z^*$ feasible by lump-sum transfer for e.

The next theorem shows that the impossibility persists with **Laissez-faire**^{W} if one looks for orderings that satisfy the widely-used **Weak Pareto** principle.[10] The proof builds on the same intuitions as Theorem 1 and it is relegated to the Appendix.

Theorem 2. *There does not exist any* ***R(e)*** *that satisfies* ***Weak Pareto, Laissez-Faire***^{W}, ***Same-Preference Anonymity*** *and* ***Fleming (1952) Separability***

These impossibilities are surprising[11] because Bosmans, Decancq, and Ooghe (2018), Fleurbaey and Maniquet (2011, 2017), and Piacquadio (2017) all characterized possibilities with similar axioms.[12] The present paper only differs by allowing price dispersion in the cross-section and this is enough to create impossibilities. Indeed, price dispersion implies that budgets may be non-nested so the optimal Laissez-faire allocation in one economy is feasible in the other economy, and *vice versa*. Because **Laissez-faire** imposes a different optimal allocation in both economies but **Fleming (1952) Separability** requires a stability of the social ordering across economies, we reached impossibilities.

[10] Weak Pareto imposes that $\forall e \in \mathcal{E}$ and z, z' such that $z_i \succ z_i'$ for all $i \in \mathcal{N}$, then z**P(e)**z'.

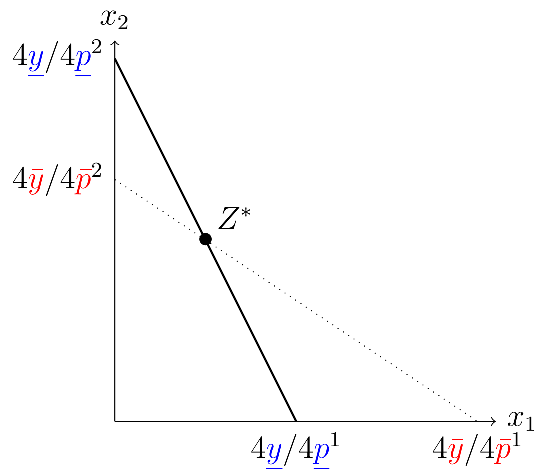
[11] I note that Bosmans and Öztürk (2022) studied the impossibility between a laissez-faire principle and the Pareto principle. Yet, their laissez-faire principle is much stronger than ours as it holds under any initial distribution of endowments while ours restrict it to economies with $(\omega_i, p_i) = (\omega, p)$ for all $i \in \mathcal{N}$. As such, their axiom is closer to libertarianism while the present axiom is closer to the ethics of equality of opportunity.

[12] It is not difficult to see from the proof of Theorem 1 that the impossibilities would persist if we replace **Same-Preference Anonymity** with the transfer-based axiom of Compensation considered in Bosmans, Decancq, and Ooghe (2018).

4 Possibilities

In the previous section, we constructed two economies in which all agents face the same budget within each economy, i.e. two equal-budget economies. The Laissez-faire allocations in these two economies generate the same aggregate bundle $z_1^* + z_1^* + z_3^* + z_3^* = Z^* = z_2^* + z_2^* + z_4^* + z_4^*$, yet the aggregate budget sets, depicted in Figure 2, are starkly different.

Figure 2: Aggregate budgets in the proof of Theorem 1.



The quantity of good 1 is denoted by x_1 on the horizontal axis and the quantity of good 2 is denoted by x_2 on the vertical axis. The consumption vector $Z^* \in \mathbb{R}_+^L$ is the total quantity of resources used in each Laissez-faire allocation, i.e. $z_1^* + z_1^* + z_3^* + z_3^* = Z^* = z_2^* + z_2^* + z_4^* + z_4^*$.

Because the **Laissez-faire** axiom induces a different ranking of allocations in these two economies, but repeated applications of **Fleming (1952) Separability** impose a stability of the ordering, we reached an impossibility. Arguably, Separability axiom is too strong in this context: the social ordering should not be completely immune to changes in indifferent agents' characteristics, because they shape what is feasible from lump-sum transfers, and in turn affect how **Laissez-faire** axioms bite in equal-budget economies. At minimum, the weakening strategy must forbid applications of Separability between equal-budget economies whose aggregate budgets are not nested.

The natural way would be to impose Separability between two generic economies e and e' only if the aggregate budget of one is nested into the other. However, this is not enough. Indeed, imagine $\mathcal{B}(e)$ is nested in $\mathcal{B}(e')$ and $\mathcal{B}(e'')$ is nested in $\mathcal{B}(e')$. By repeated applications of Separability, we will be able to compare $\mathcal{B}(e)$ and $\mathcal{B}(e'')$, yet they may be non-nested simplices. To avoid this, we must ensure that chained applications of Separability only allow us to reach one and only one equal-budget economy.

Our weakening strategy will consist of applying Separability between two economies only if they share the same underlying equal-budget economy $L(e') = L(e)$. We need to choose what *underlying* means, that is the shape of the $L(\cdot)$ function. We select it so that the equal-budget economy is as close as possible to the generic economy e. More precisely, the equal-budget economy $L(e)$ is the largest (with respect to set inclusion) equal-budget economy nested in the aggregate budget of e.

$$L(e) = \max_{\subseteq} \mathcal{B}\big(\{(\succsim_i, \omega, p)_{i\in\mathcal{N}}\}\big) \subseteq \mathcal{B}(e).$$

This definition ensures that the set of allocations that can be reached by lump-sum transfers from the equal-budget economy is as close as possible to the set of allocations that can be reached by lump-sum transfers from the generic economy e. We are now equipped to define the logical weakening of **Fleming (1952) Separability**. **No-Growth Separability** applies only between economies that share the same $L(e)$.

No-Growth Separability: $\forall e, e' \in \mathcal{E}$ and the partition $\mathcal{K} + \mathcal{M} = \mathcal{N}$ with $e_{\mathcal{K}} = \{(\succsim_i, \omega_i, p_i)_{i\in\mathcal{K}}\}$, $e'_{\mathcal{K}} = \{(\succsim'_i, \omega'_i, p'_i)_{i\in\mathcal{K}}\}$, $e_{\mathcal{M}} = \{(\succsim_i, \omega_i, p_i)_{i\in\mathcal{M}}\} = e'_{\mathcal{M}}$, if $L(e) = L(e')$, then

$$(z_{\mathcal{K}}, z_{\mathcal{M}})\mathbf{R}(\mathbf{e}_{\mathcal{K}},\mathbf{e}_{\mathcal{M}})(z_{\mathcal{K}}, z'_{\mathcal{M}}) \implies (z'_{\mathcal{K}}, z_{\mathcal{M}})\mathbf{R}(\mathbf{e}'_{\mathcal{K}},\mathbf{e}_{\mathcal{M}})(z'_{\mathcal{K}}, z'_{\mathcal{M}})$$

This breaks the impossibility of Theorem 1 and the comparison of non-nested simplices depicted in Figure 2: chained applications of Separability can lead to one and only one equal-budget economy. It will be useful to understand the geometric consequences of imposing **No-Growth Separability**. They are captured in the following lemma.

Lemma 1. *Any two economies e and e' are such that $L(e) = L(e')$ if and only if*

$$\textit{for each } l = \{1, ..., L\} : \sum_{i\in\mathcal{N}} \frac{y_i}{p_i^l} = \sum_{i\in\mathcal{N}} \frac{y'_i}{p_i'^l}$$

Proof. Any equal-budget economy has an aggregate budget defined by $\{Z \in \mathbb{R}^L_+ : pZ \leq Ny\}$. This is a simplex whose intercept with the axis l is Ny/p^l. Because $\mathcal{B}(e)$ is convex, we have

$$\mathcal{B}\big(\{(\succsim_i, \omega, p)_{i\in\mathcal{N}}\}\big) \subseteq \mathcal{B}(e) \iff Ny/p^l \leq \sum_{i\in\mathcal{N}} \frac{y_i}{p_i^l} \text{ for each } l = \{1, ..., L\}$$

The largest equal-budget economy that satisfies the set inclusion is therefore the one where the inequality is saturated. Hence, $L(e) = L(e')$ if and only if $\sum_{i\in\mathcal{N}} \frac{y_i}{p_i^l} = \sum_{i\in\mathcal{N}} \frac{y'_i}{p_i'^l}$, which is the stated condition. ■

This lemma shows that our weakening of Separability aimed at comparing economies with similar sets of lump-sum transfer feasible allocations is equivalent to comparing economies that have identical resources, if one defines resources in good l as $\sum_{i\in\mathcal{N}} \frac{y_i}{p_i^l}$, that is the maximal quantity of good l that could ever be consumed in that economy. Geometrically, the resource in good l is the intercept of the aggregate budget with the l-axis.

The two main theorems of this paper will make use of **No-Growth Separability** as well as **Same-Preference Anonymity**. However, they will feature two logically stronger versions of the *laissez-faire* principle which I now define in turn.

First, **Egalitarian Responsibility** imposes that when all agents have equal endowments and prices, any reduction in lump-sum transfers inequality is a social improvement. Of course, the Laissez-faire allocation z^*, where no lump-sum transfers inequality remains, is still the social optimum.

Egalitarian Responsibility: $\forall e \in \mathcal{E}$ such that $(\omega_i, p_i) = (\omega, p)$ $\forall i \in \mathcal{N}$, and z, z' such that $z_k = z'_k$ $\forall k \in \mathcal{N} \backslash \{i, j\}$, and

$$
\begin{aligned}
z_i &\in \max_{\succsim_i} B(t_i, \omega, p) \qquad & z'_i &\in \max_{\succsim_i} B(t'_i, \omega, p) \\
z_j &\in \max_{\succsim_j} B(t_j, \omega, p) \qquad & z'_j &\in \max_{\succsim_j} B(t'_j, \omega, p)
\end{aligned}
$$

If $|t'_i - t'_j| \geq |t_i - t_j|$ then $z\mathbf{R(e)}z'$

Second, **Utilitarian Responsibility** imposes that when all agents have equal endowments and prices, there should be no lump-sum transfers waste: all resources at the planner's disposal should be used.

Utilitarian Responsibility: $\forall e \in \mathcal{E}$ such that $(\omega_i, p_i) = (\omega, p)$ $\forall i \in \mathcal{N}$, and z, z' such that $z_k = z'_k$ $\forall k \in \mathcal{N} \backslash \{i, j\}$, and

$$
\begin{aligned}
z_i &\in \max_{\succsim_i} B(t_i, \omega, p) \qquad & z'_i &\in \max_{\succsim_i} B(t'_i, \omega, p) \\
z_j &\in \max_{\succsim_j} B(t_j, \omega, p) \qquad & z'_j &\in \max_{\succsim_j} B(t'_j, \omega, p)
\end{aligned}
$$

If $t_i + t_j \geq t'_i + t'_j$ then $z\mathbf{R(e)}z'$.

It is important to observe that both **Egalitarian Responsibility** and **Utilitarian Responsibility** imply **Laissez-faire**^{W}. Indeed, the *laissez-faire* allocation is such that there is no lump-sum transfer inequality among agents, which **Egalitarian Responsibility** finds most desirable. However, the *laissez-faire* allocation is also the one where the sum of lump-sum transfers is 0, which is preferred to all other feasible allocations (i.e. $\sum t_i \leq 0$) by **Utilitarian Responsibility**. Both axioms confirm the supremacy of *laissez-faire* allocation in equal-budget economies, but for different reasons. They also have bite beyond that as any change toward the *laissez-faire* allocation is desirable, thus being logically stronger than **Laissez-faire**^{W}.

I now present the main theorems of the paper. They show that using **No-Growth Separability** escapes the impossibility. These theorems characterize partially social welfare because they apply only when a pair of agents have unequal consumption. To fully characterize the social welfare function, we could use known axiomatic results of aggregations (see Bosmans, Decancq, and Ooghe (2018), Fleurbaey and Maniquet (2011), and Piacquadio (2017) for discussions). However, this would only affect the aggregation of welfare, but does not affect the individual welfare measure. Because only the individual welfare measure is used in applications below, I keep these partial characterizations which only require three axioms to determine individual welfare.

Both theorems pin down one possibility, and each one uses the same measure of individual consumer welfare, but they differ in their aggregation rule. The individual consumer welfare is the expenditure function with a common reference price for all consumers. Formally it is defined as

$$W_i(\tilde{p}, z_i) = \min_{z_0} \tilde{p} \cdot z_0 \text{ such that } z_0 \succsim_i z_i \text{ and } \tilde{p}^l = \frac{\sum_{i \in \mathcal{N}} y_i}{\sum_{i \in \mathcal{N}} \frac{y_i}{p_i^l}} \quad \text{for } l = \{1, ..., L\}$$

Let me stress that $W_i(\cdot)$ represents agent i's preferences: a preferred bundle increases $W_i(\cdot)$. Moreover, it is also money-metric because it is expressed in the same unit as income. I provide a graphical illustration of the construction of $W_i(\tilde{p}, z_i)$ for a two-agent case with two goods in Figure 3. In panel (a), I show that $\tilde{p}^l$ is the price that renders the total resources of the economy affordable with the aggregate income $\sum_i y_i$. In panel (b), I show that the welfare of agent i consuming z_i is the minimal expenditures at price $\tilde{p}$ required for agent i to be indifferent to z_i.

Theorem 3 shows that **Egalitarian Responsibility**, combined with **Same-Preference Anonymity** and **No-Growth Separability** leads to a social welfare function that has inequality aversion in that money-metric utility.

Theorem 3. *For $N \geq 5$, **R(e)** satisfies **No-Growth Separability**, **Same-Preference Anonymity** and **Egalitarian Responsibility** if and only if for z, z' and e such that*

$$W_i(\tilde{p}, z_i') \leq W_i(\tilde{p}, z_i) \leq W_j(\tilde{p}, z_j) \leq W_j(\tilde{p}, z_j')$$

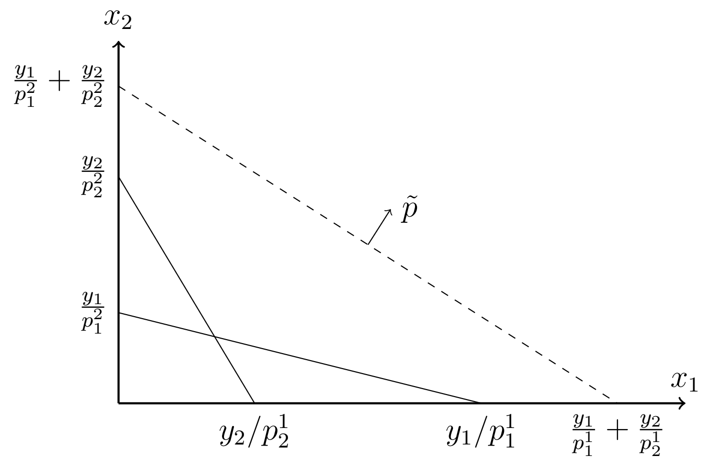
*while $z_k' = z_k$ for all $k \in \mathcal{N} \backslash \{i, j\}$, then z**R(e)**z'.*

The proof is relegated to the appendix and its strategy is as follows. First, by **Egalitarian Responsibility**, the premise of the theorem implies that z is preferred to z' when all agents have identical budgets. Then, repeated applications of **No-Growth Separability** and **Same-Preference Anonymity** allows us to extend that ranking to any arbitrary economy e, provided we have five agents in the economy.

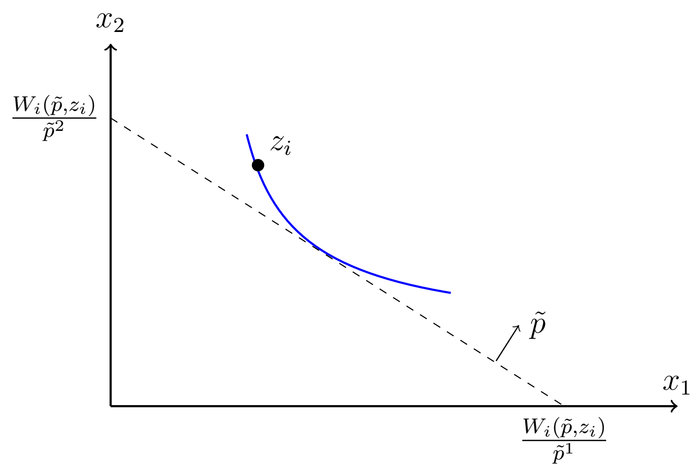
This theorem shows that the social welfare function has an infinite aversion to inequality in the particular money-metric utility $W_i(\tilde{p}, z_i)$. However, observe that not all consumption inequalities are deemed unfair. Indeed, unequal preferences imply that agents consume different bundles in the Laissez-faire allocation even if they have the same budgets. Yet, this allocation is the optimal one with respect to Theorem 3. This is why these axioms reflect the ethics of equality of *opportunity* and not of *outcomes*: budget sets should be equalized but not consumption vectors.

The next theorem shows the consequences of replacing **Egalitarian Responsibility** by **Utilitarian Responsibility**. Its proof is relegated to the appendix as well.

Figure 3: Illustrations of the construction of the interpersonal consumer welfare measure.



(a) Illustration of the construction of $\tilde{p}$.



(b) Illustration of the construction of $W_i(\tilde{p}, z_i)$.

Panel (a) shows that $\tilde{p}$ is the common price vector that makes the total quantity of resources in the economy affordable with the aggregate income $y_1 + y_2$. Panel (b) shows that the welfare $W_i(\tilde{p}, z_i)$ of agent i consuming z_i is the amount of expenditures at price $\tilde{p}$ needed to be indifferent to z_i.

Theorem 4. *For $N \geq 5$, **R(e)** satisfies **No-Growth Separability**, **Same-Preference Anonymity** and **Utilitarian Responsibility** if and only if for z, z' and e such that*

$$W_i(\tilde{p}, z_i) + W_j(\tilde{p}, z_j) \geq W_i(\tilde{p}, z'_i) + W_j(\tilde{p}, z'_j)$$

*while $z_k = z'_k \; \forall k \in \mathcal{N} \backslash \{i, j\}$, then z**R(e)**z'.*

These results deserve some comments. First, these two theorems show that there exist social welfare functions endogenized by the axioms. They differ in the aggregator used to compute social welfare (the *maximin* and the *sum*, respectively), but they use the very same measure of interpersonal welfare, i.e. $W_i(\tilde{p}, z_i)$. It suggests that this money-metric utility function should be used to compare the welfare of heterogeneous consumers if one abides by **Same-Preference Anonymity**, **Laissez-faire**^{W} and **No-Growth Separability**, while the choice of the aggregator depends on which logical strengthening the ethical observer prefers between **Utilitarian Responsibility** and **Egalitarian Responsibility**. Whether the observer should use the sum or the maximin to assess collective welfare changes in any economy can be reduced to the problem of evaluating collective welfare between equally-endowed individuals. If the planner wants to minimize welfare inequalities among same-budget individuals, then the maximin should be adopted as social welfare aggregator. If the observer rather wishes to maximize the total welfare of same-budget individuals, then it should use the summation.

Second, this welfare measure only uses ordinal and non-comparable elements of preferences to build a welfare measure. Indeed, the only component of preferences needed to compute $W_i(\tilde{p}, z_i)$ is the indifference curve at z_i, not the utility level. Despite that, this measure allows interpersonal comparisons of welfare between individuals. Thus, this measure is in line with the long-standing tradition since Robbins (1938) and Samuelson (1947) of not comparing cardinal utility levels across individuals.[13] Besides, both measures also satisfy the Pareto principle.

Third, that the sum of expenditure functions is an attractive way of measuring welfare has a long history in welfare economics marked with controversies over the choice of the reference price vector. First, the Hicksian compensating and equivalent variations are expenditure functions using post- and pre-reform prices as reference situation respectively[14], but a social welfare function using their sum may lead to intransitive assessment as is known since Boadway (1974) and Scitovsky (1941).[15] Second, money-metric utility functions have been criticized by Blackorby and Donaldson (1988) because they may yield anti-redistributive consequences for some reference prices[16] even when the social welfare function is egalitarian. Third, Blackorby, Laisney, and Schmachtenberg (1993) and Roberts (1980) underlined the arbitrariness of the choice of the reference price and studied price-independent welfare assessment. What the present paper offers are social welfare functions based on money-metric utility that overcome each of these critiques: they are transitive, allow for inequality aversion, and their reference situation is endogenized by the axioms. Moreover, $\tilde{p}$ is independent of the normalization of the numéraire of the economy. I now present some consequences for applied welfare analysis.

[13]This is not a new result in itself, as it is characteristic of the fairness literature in social choice. See Bosmans, Decancq, and Ooghe (2018), Fleurbaey (2003), Fleurbaey and Maniquet (2011), and Piacquadio (2017) among others.

[14]The widely-used consumer surplus is a Hicksian variation when preferences are quasilinear. See Hicks (1939, 1941, 1942, 1943, 1945) for the original contributions.

[15]See Blackorby and Donaldson (1990) for a review.

[16]Bosmans, Decancq, and Ooghe (2018), Fleurbaey and Maniquet (2011), and Schlee and Khan (2022, 2023) discussed several ways to escape this critique.

5 Applications

The following applications all use the interpersonal consumer welfare measure $W_i(\cdot)$ and are valid whether Theorems 3 or 4 are used, that is whether the aggregator of social welfare is the minimum or the sum. The fact that the axioms endogenize the same reference price $\tilde{p}$ allows to contribute to various applied welfare problems without taking a stance on which aggregator should be used.

5.1 Income of nations

In the index number problem[17] initiated by Fisher (1922), the economist must compare the *real income* of a set of countries based on a dataset of observed prices p_i and quantities consumed z_i and decompose this real income into a price index and a quantity index. Our analysis of welfare above can readily be applied to this problem if each country is considered as an agent and *real income* is considered as the expenditure function $W_i(\tilde{p}, z_i)$. More formally, real income can be decomposed into a price index P_i and a quantity index $Q_{i,j}$ as follows

$$P_i = \frac{W_i(p_i, z_i)}{W_i(\tilde{p}, z_i)}$$
$$Q_{i,j} = \frac{W_i(\tilde{p}, z_i)}{W_j(\tilde{p}, z_j)}$$

where the price index P_i reflects the deflation needed to compare the prices faced by country i and $Q_{i,j}$ captures the real income difference between countries i and j.[18]

The present paper argues for a common price $\tilde{p}$ to deflate countries' consumption. Interestingly, the widely-used Geary (1958) method, which underpins the Penn World Table and the international dollar, also computes a world price[19] $\mathring{p}^l$ for each good l defined as

$$\mathring{p}^l = \frac{\sum_i p_i^l z_i^l \epsilon_i}{\sum_i z_i^l}$$

where ϵ_i is a normalization of countries' income such that $\epsilon_i = \frac{\sum_l \mathring{p}^l z_i^l}{\sum_l p_i^l z_i^l}$. The real income of country i is then computed as the cost of that country's bundle z_i at the world prices $\mathring{p}$. While this approach ignores preferences, Neary (2004) shows that consistency with a representative agent preferences can be obtained by setting ϵ_i as the ratio of expenditure functions at

[17] That expenditure functions and welfare economics are useful for index number theory is known at least since the classical contributions of Deaton (1979) and Samuelson and Swamy (1974).

[18] In the literature, the former is a true Könus price index while the latter is an Allen true quantity index. This paper's measure escapes the Van Veelen (2002) impossibility by using third-country information when computing the real income of a pair of countries.

[19] Deaton and Heston (2010) argued that the main advantage of the approach is the preservation of integrability. Indeed, because it uses a single world price for each good, a summation across goods does not perturb the ranking.

these prices.

A necessary condition for $\tilde{p}$ to be identical to $\mathring{p}$ is that, for each good l, expenditure shares $\frac{p_i^l z_i^l}{y_i}$ are identical across countries. This is quite unlikely to hold in any international comparisons: we know since Engel (1857) that richer countries spend proportionally less on food. As a consequence, we should expect our modified PPP to yield different empirical results than the conventional Geary method.

Moreover, observe that if there is a preference shock to one country while all prices and income in the world are constant, Geary's prices $\mathring{p}^l$ will change, while this paper's $\tilde{p}$ will not. Indeed, the former depends on expenditure shares of each country, while the latter only depends on prices and income. Because of this, the deflation proposed in this paper is robust to preference heterogeneity. Importantly, this modification of the purchasing power parity does not require more data than what is commonly used.

5.2 Lifetime inflation and growth

We can also interpret the standard consumption space over time by considering each good l is a time period. For clarity, let me substitute the superscript l by $t = \{1, ..., T\}$. The problem for the growth economist consists in measuring inflation and real income growth over the past T periods for a set of countries based on a dataset of country i's prices p_i^t and consumption z_i^t. Preferences are defined over the T periods, i.e. over the stream of consumption z_i, and must be understood as *lifetime* welfare.

The present paper suggests that the welfare-relevant inflation and growth indices between consumers i and j over the T periods are

$$\Pi_i = \frac{W_i(p_i, z_i)}{W_i(\tilde{p}, z_i)}$$
$$Q_{i,j} = \frac{W_i(\tilde{p}, z_i)}{W_j(\tilde{p}, z_j)}$$

Interestingly, in such case the reference price vector $\tilde{p}$ becomes close to the GDP deflator i.e. $\tilde{p}^t = \frac{\sum_i y_i}{\sum_i y_i / p_i^t}$.[20] Unlike the Consumer Price Index (CPI), the GDP deflator does not depend on the weights of each consumption good in the bundle consumed, but rather only on income and prices. Again, this can be directly traced back to the axioms: **Laissez-faire** precludes the economist from choosing a particular consumption good like the CPI basket, because it would give prominence to a particular kind of preferences while all preferences should be deemed equal.

Importantly, this does not mean that preferences are ignored. Indeed, Π_i and $Q_{i,j}$ both depend on preferences as they use expenditure functions as arguments. However, the measuring rod does not: $\tilde{p}$ only depends on consumption possibilities.

[20] In practice, the GDP deflator is computed on more goods than mere private consumption goods and include the price of investments as well as government expenditures.

5.3 Year-by-year inflation and growth

Consider the case where each time period the preferences of the agent under study may vary. In such case, we can consider that the index i used above may be substituted by t. In turn, measuring welfare interpersonally becomes a measure of year-by-year inflation and welfare growth for a single country when prices, endowments, and preferences may change from one year to the next. The relevant inflation and growth indices are

$$\Pi_t = \frac{W_t(\tilde{p}, z_t)}{W_t(p_t, z_t)}$$
$$Q_{t,t-1} = \frac{W_t(\tilde{p}, z_t)}{W_{t-1}(\tilde{p}, z_{t-1})}$$

In this setup, the reference price for a good l $\tilde{p}^l$ becomes the ratio between the sum of expenditures over time over the the sum of maximal quantity of good l that could have been consumed.

$$\tilde{p}^l = \frac{\sum_t y_t}{\sum_t y_t / p_t^l}$$

While traditional inflation and growth measures typically ignore preferences or assumed homothetic preferences, a recent contribution by Jaravel and Lashkari (2024) provides an algorithm to measure inflation and growth for non-homothetic and general preferences. Interestingly, they also rely on expenditure functions with a common, time-invariant reference price vector which they call *base period*. However, neither their theory nor their algorithm provides guidance on which base period to choose. While their estimates significantly differ from standard estimates derived under homothetic preferences, the magnitude of these differences depends on the *base period*. What the present paper gives is a normative justification for the base period that connects the measure with consumer welfare.

6 Conclusion

This paper showed that, once heterogeneous consumers face different prices, interpersonal comparison, deflation, and aggregation can be jointly determined. If consumption inequalities due to heterogeneous preferences do not hurt social welfare, same-preference consumers are treated anonymously, and social welfare is consistent across economies, then individual consumer welfare is evaluated by the expenditure required to reach her indifference curve at a common price vector. We showed that this common price vector is the income-weighted harmonic mean of individual prices, and it can be retrieved without additional data in various applications.

The broader goal of the paper was to show that statistics on the costs and standards of living embed normative choices about how heterogeneous consumers should be compared. By linking the current practice of statistical offices with welfare economics, the paper hopes to make these choices more explicit.

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A Proofs

Proof of Theorem 2. I proceed again by contradiction. Consider the same setup as in the proof of Theorem 1 and z_5, z_6 such that (z_5, z_5, z_6, z_6) is feasible as lump-sum transfers from the economy where all agents have $B(0, \underline{\omega}, \underline{p})$ while $z_5 \succ_i z_1^*$ while $z_6 \succ_j z_3^*$. This is illustrated in Figure A.1.

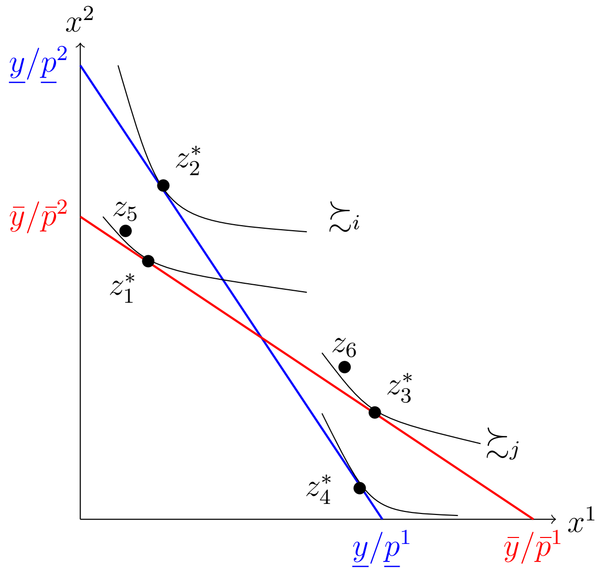
Using the exact same steps as in Theorem 1 but with a weak preference relation rather than a strict one, we get that by ***Laissez-Faire***^{W}, ***Same-Preference Anonymity***, Transitivity and **Fleming (1952) Separability**,

$$(z_2^*, z_1^*, z_3^*, z_4^*) \qquad \mathbf{R}\left(\left\{(\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p})\right\}\right) \qquad (z_2^*, z_2^*, z_4^*, z_4^*) \tag{A.1}$$

By **Fleming (1952) Separability**,

$$(z_1^*, z_1^*, z_3^*, z_3^*) \qquad \mathbf{R}\left(\left\{(\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p})\right\}\right) \qquad (z_1^*, z_2^*, z_4^*, z_3^*)$$

Figure A.1: Illustration of the proof of Theorem 2.



Notes: The quantity of good 1 is denoted by x_1 on the horizontal axis and the quantity of good 2 is denoted by x_2 on the vertical axis. The economies have been chosen such that the Laissez-faire allocation in the blue economy is feasible by lump sum transfers from the red economy, and *vice versa*, that is $\frac{z_1+z_3}{2} = \frac{z_2+z_4}{2}$. Moreover, (z_5, z_6, z_5, z_6) is feasible by lump sum transfers from the when all agents have $B(0, \underline{\omega}, \underline{p})$.

By **Same-Preference Anonymity** and Transitivity,

$$(z_1^*, z_1^*, z_3^*, z_3^*) \qquad \mathbf{R}\left(\left\{(\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p})\right\}\right) \qquad (z_2^*, z_1^*, z_3^*, z_4^*)$$

By Transitivity with equation A.1,

$$(z_1^*, z_1^*, z_3^*, z_3^*) \qquad \mathbf{R}\left(\left\{(\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p})\right\}\right) \qquad (z_2^*, z_2^*, z_4^*, z_4^*)$$

By **Laissez-Faire**^{W},

$$(z_2^*, z_2^*, z_4^*, z_4^*) \qquad \mathbf{R}\left(\left\{(\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p})\right\}\right) \qquad (z_5, z_5, z_6, z_6)$$

By Transitivity,

$$(z_1^*, z_1^*, z_3^*, z_3^*) \qquad \mathbf{R}\left(\left\{(\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p})\right\}\right) \qquad (z_5, z_5, z_6, z_6)$$

which contradicts **Weak Pareto** and completes the proof. ■

Proof of Theorem 3. The proof of the "if" proceeds by construction. Take a generic economy $e \in \mathcal{E}$ and two allocations z, z' such that the premise holds. Without loss of generality, set $i = 3$ and $j = 4$.

By **Egalitarian *Responsibility***, we have

$$z \ \ \mathbf{R}\left(\left\{(\succsim_1, \tilde{\omega}, \tilde{p})(\succsim_2, \tilde{\omega}, \tilde{p})(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_i, \tilde{\omega}, \tilde{p})_{i\in\{5,\ldots N\}}\right\}\right)(z_1, z_2, z_3', z_4', \ldots z_N)$$

By ***No-Growth Separability***, we can write

$$z \ \ \mathbf{R}\left(\left\{(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_i, \tilde{\omega}, \tilde{p})_{i\in\{5,\ldots N\}}\right\}\right)(z_1, z_2, z_3', z_4' \ldots z_N)$$

By ***Same-Preference Anonymity*** applied twice,

$$(z_3', z_4', z_1, z_2 \ldots, z_N)\mathbf{I}\left(\left\{(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_i, \tilde{\omega}, \tilde{p})_{i\in\{5,\ldots N\}}\right\}\right)(z_1, z_2, z_3', z_4' \ldots, z_N)$$

By transitivity,

$$z \ \ \mathbf{R}\left(\left\{(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_i, \tilde{\omega}, \tilde{p})_{i\in\{5,\ldots N\}}\right\}\right)(z_3', z_4', z_1, z_2 \ldots z_N)$$

By ***Same-Preference Anonymity*** applied twice,

$$z \ \ \mathbf{I}\left(\left\{(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_i, \tilde{\omega}, \tilde{p})_{i\in\{5,\ldots N\}}\right\}\right)(z_3, z_4, z_1, z_2 \ldots, z_N)$$

By transitivity,

$$(z_3, z_4, z_1, z_2 \ldots z_N)\mathbf{R}\left(\left\{(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_i, \tilde{\omega}, \tilde{p})_{i\in\{5,\ldots N\}}\right\}\right)(z_3', z_4', z_1, z_2 \ldots z_N)$$

By ***No-Growth Separability***,

$$(z_3, z_4, z_1, z_2 \ldots, z_N)\mathbf{R}\left(\left\{(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_k, y_k, p_k)_{k\in\{3,4\}}(\succsim_5, \hat{\omega}_5, \hat{p})(\succsim_i, \tilde{\omega}, \tilde{p})_{i\in\{6,\ldots N\}}\right\}\right)(z_3', z_4', z_1, z_2 \ldots, z_N)$$

only if we can find $\hat{\omega}_5$ and $\hat{p}$ such that

$$\text{for each } l = \{1, \ldots, L\} : 3\frac{\tilde{y}}{\tilde{p}^l} = \frac{y_3}{p_3^l} + \frac{y_4}{p_4^l} + \frac{\hat{p}\hat{\omega}_5}{\hat{p}^l} \tag{A.2}$$

By ***Same-Preference Anonymity*** applied four times and transitivity,

$$z \ \ \mathbf{R}\left(\left\{(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_k, \omega_k, p_k)_{k\in\{3,4\}}(\succsim_5, \hat{\omega}_5, \hat{p})(\succsim_i, \tilde{\omega}, \tilde{p})_{i\in\{6,\ldots N\}}\right\}\right)(z_1, z_2, z_3', z_4' \ldots z_N)$$

By ***No-Growth Separability***,

$$z \ \ \mathbf{R(e)}(z_1, z_2, z_3', z_4' \ldots, z_N)$$

if and only if

$$\text{for each } l = \{1, ..., L\} : (N-3)\frac{\tilde{y}}{\tilde{p}^l} + \frac{\hat{p}\hat{\omega}_5}{\hat{p}^l} = \sum_{i \in \mathcal{N}\setminus\{3,4\}} \frac{y_i}{p_i^l} \tag{A.3}$$

Summing over equations (A.2) and (A.3), we get

$$\text{for each } l = \{1, ..., L\} : N\frac{\tilde{y}}{\tilde{p}^l} + \frac{\hat{p}\hat{\omega}_5}{\hat{p}^l} = \sum_{i \in \mathcal{N}} \frac{y_i}{p_i^l} + \frac{\hat{p}\hat{\omega}_5}{\hat{p}^l}$$

which holds by construction of $\tilde{\omega}$ and $\tilde{p}$ and Lemma 1. The result is obtained by observing that $(z_1, z_2, z_3', z_4'..., z_N)$ is the same allocation as z', proving z**R(e)**z', the required result.

To prove the "only if", observe that $W_i(\tilde{p}, z_i)$ does not depend on the individual's endowments (ω_i, p_i) such that a permutation between equal-preference individuals leads the ordering unchanged, proving that **R(e)** satisfies **Same-Preference Anonymity**.
As for **No-Growth Separability**, observe that any change of endowments such that $\sum_i \frac{y_i}{p_i^l}$ is unchanged for all l does not modify $\tilde{\omega}$ nor $\tilde{p}$. The proof for **Egalitarian Responsibility** is trivial and left to the reader. ∎

Proof of Theorem 4. Without loss of generality, consider that agents 3 and 4 are i and j. Then, by ***Utilitarian Responsibility***, we have

$$z \;\; \mathbf{R}\left(\left\{(\succsim_1, \tilde{y}, \tilde{p})(\succsim_2, \tilde{y}, \tilde{p})(\succsim_3, \tilde{y}, \tilde{p})(\succsim_4, \tilde{y}, \tilde{p})(\succsim_i, \tilde{y}, \tilde{p})_{i\in\{5,...N\}}\right\}\right)(z_1, z_2, z_3', z_4', ...z_N)$$

The remaining steps of the proof follow exactly the proof of the previous theorem. ∎